

AI VIDEO ANALYSIS AND FAKE DETECTION SYSTEM USING WIND

A PROJECT REPORT

Submitted By

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BONAFIDE CERTIFICATE

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VIVA VOCE EXAMINATION

The viva-voce examination of the project work titled “**AI VIDEO ANALYSIS AND FAKE DETECTION SYSTEM USING WIND**” submitted by **JATHIKA KANNAN (110822104034)** was held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The proliferation of misinformation and manipulated video content across digital platforms has emerged as a critical societal challenge in the modern information age. This project presents a comprehensive AI-driven Video Analysis and Fake Detection System that integrates advanced machine learning, natural language processing, and speech recognition technologies to automatically detect fraudulent video content and identify misleading narratives. The system employs the WIND (Whisper-based Integrated Neural Decoder) speech recognition model combined with sophisticated video processing algorithms to extract, analyze, and authenticate video content through multiple parallel detection mechanisms.

The proposed system implements a complete pipeline architecture that processes user-uploaded videos through sequential stages of quality assessment, frame extraction, face detection, audio analysis, speech transcription, and fake news detection. The backend infrastructure utilizes Django framework, OpenCV for computer vision tasks, Firebase for secure authentication, and Cloudinary for efficient cloud storage. The system generates comprehensive analytics including video quality metrics, transcript accuracy rates (Word Error Rate), and fake news probability scores, presented through an interactive dashboard interface.

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LIST OF SYMBOLS, ABBREVIATIONS AND DEFINITIONS

SYMBOLS



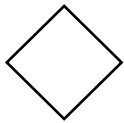
: External Entity



: Use Case



: Flow of Data



: Conditions



Actor

: Actor

ABBREVIATIONS

NLP : Natural language processing

WER : Word Error Rate

WIND : Windowed Interface

FPS : Frame Per Second

WAV : Waveform Audio File Format

IDE : Integrated Development Environment

GPU : Graphics Processing Unit

API : Application Programming Interface

UI : User Interface

HTTPS : Hypertext Transfer Protocol Secure

DEFINITIONS

1. Video Analysis

Video analysis is the process of extracting meaningful information from video frames. In this project, it includes analyzing video quality, detecting faces, and identifying suspicious or manipulated content.

2. Fake Video Detection

Fake video detection refers to identifying manipulated or misleading videos using artificial intelligence techniques. The system analyzes video frames and speech to determine authenticity.

3. WIND (Windowed Inference)

WIND is an algorithm used to improve speech recognition accuracy by analyzing audio signals in small time windows and identifying meaningful speech segments.

4. Face Detection

Face detection is a computer vision technique used to locate and identify human faces within video frames using algorithms such as Haar Cascades or deep learning models.

5. Speech Recognition

Speech recognition is the technology that converts spoken language from video audio into text. This project uses AI models to generate transcripts from video audio.

6. Word Error Rate (WER)

Word Error Rate is a metric used to evaluate the accuracy of speech recognition systems by comparing the predicted transcript with the actual reference text.

7. Window Frame Analysis

Window frame analysis divides the video into small segments or windows and analyzes each segment individually to detect anomalies or Dashboard analytics provides visual

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

1.1 SYSTEM OVERVIEW

The **AI Video Analysis and Fake Detection System using WIND** is an intelligent software platform designed to analyze video content and identify potentially manipulated or fake videos using artificial intelligence techniques. The system integrates multiple technologies such as computer vision, speech recognition, and machine learning to examine different components of a video. By processing both visual and audio information, the system provides a comprehensive analysis that helps determine the authenticity and quality of uploaded video files.

The system works by allowing users to upload video files through a web-based interface developed using the Django framework. Once the video is uploaded, the system performs several automated processes including video validation, metadata extraction, video quality assessment, face detection, and audio transcription. The video is divided into frames for analysis, where parameters such as blur, brightness, motion, and noise are evaluated. Additionally, speech recognition models are used to convert the audio content into text, which is further analyzed to detect misleading or fake information.

1.2 OBJECTIVE

The main objective of the **AI Video Analysis and Fake Detection System using WIND** is to develop an intelligent system capable of automatically analyzing video content and identifying fake or manipulated videos using artificial intelligence techniques. The project aims to examine both visual and audio components of a video to ensure accurate detection of misleading or altered media. The system performs video quality assessment by evaluating factors such as blur, brightness, motion, and noise to determine the reliability of the video content. Another objective is to implement speech recognition technology to convert the audio in videos into text for further analysis. Additionally, the project integrates face detection techniques to identify human faces in video frames, which helps in verifying authenticity. The system also focuses on generating automated reports that summarize video analysis results and fake detection outcomes. Furthermore, a user-friendly web interface is developed using the Django framework to allow users to easily upload videos and view analysis results efficiently. Furthermore, the system aims to automatically generate **comprehensive analysis reports** that

summarize video quality metrics, face detection results, transcription output, and fake detection predictions. These reports help users clearly understand the analysis results without needing technical expertise. Another objective is to design a **user-friendly web interface** using the Django framework, allowing users to easily upload video files and access analysis results through an interactive dashboard.

Finally, the project seeks to improve the **accuracy, efficiency, and reliability of fake video detection systems** by combining computer vision, natural language processing, and machine learning technologies. By integrating multiple analysis techniques into a single platform, the system provides a comprehensive solution for identifying fake or manipulated videos and supporting digital media verification. This contributes to reducing misinformation and improving trust in online multimedia content.

1.3 SYSTEM STUDY

1.3.1 Existing System

The existing systems for video verification and fake content detection mainly rely on manual analysis or limited automated tools. In many cases, users must manually inspect videos to determine whether the content is authentic or manipulated. This process is time-consuming, inefficient, and often inaccurate, especially when dealing with a large number of videos shared across social media platforms. Traditional systems typically focus on only one aspect of analysis, such as image verification or audio detection, without combining multiple techniques to perform a comprehensive evaluation of video content.

Some currently available tools use basic machine learning or computer vision algorithms to detect manipulated videos, but these systems often have limitations in accuracy and scalability. Many existing solutions analyze only visual features of a video and ignore the audio component, which may contain important clues related to misinformation. Additionally, these systems may not perform detailed quality assessment or frame-level analysis, which makes it difficult to detect subtle manipulations such as deepfakes or edited segments within a video.

Another limitation of existing systems is the lack of integrated platforms that combine video quality analysis, face detection, speech recognition, and fake detection in a single framework. Many tools also do not provide user-friendly interfaces or automated reporting features, making them difficult for non-technical users to operate. Due to these challenges, existing systems are not fully capable of efficiently detecting fake or misleading videos in modern digital environments, highlighting the need for a more advanced and integrated solution.

1.3.2 Literature Survey

The rapid growth of digital media and social networking platforms has increased the spread of manipulated or fake videos, creating serious challenges for verifying the authenticity of multimedia content. Researchers have proposed various artificial intelligence and machine learning techniques to detect fake videos and deepfakes. Many studies focus on using computer vision methods to analyze facial movements, frame inconsistencies, and visual artifacts in videos. Deep learning models such as convolutional neural networks (CNNs) have been widely used to detect facial manipulation and identify patterns that indicate video tampering. These techniques have shown promising results in identifying altered visual content.

Advantages:

- Existing systems provide automated tools for detecting manipulated or deepfake videos.
- Machine learning and deep learning algorithms improve detection accuracy compared to manual verification.
- Computer vision techniques allow systems to analyze facial expressions and frame patterns effectively.
- Speech recognition technologies help convert audio from videos into text for further analysis.
- Some tools can process large volumes of video data efficiently

Disadvantages:

- Many systems focus only on visual analysis and ignore audio or speech information. the analysis process, and view the results in an organized manner. The interface is developed using modern web technologies.

- The system integrates advanced technologies such as artificial intelligence, computer vision.

The system integrates advanced technologies such as artificial intelligence, computer vision, and speech recognition to perform comprehensive analysis of both visual and audio components of a video.

- Some detection methods require high computational power and advanced hardware.

- Accuracy may decrease when videos are heavily edited or compressed.

- Existing tools often lack integrated platforms that combine multiple analysis methods.

- Many systems do not provide user-friendly interfaces for non-technical users.

Another area of research focuses on audio and speech analysis for detecting misinformation in video content. Speech recognition systems are used to convert spoken language in videos into text transcripts, which can then be analyzed using natural language processing techniques. Researchers have also introduced methods like **Windowed Inference (WIND)** to improve the accuracy of speech recognition models by analyzing audio signals in smaller time windows. Combining visual analysis with audio transcription allows systems to detect inconsistencies between video content and spoken information.

Despite these advancements, many existing studies focus only on a single type of analysis such as facial detection or speech recognition. Few systems combine video quality assessment, frame analysis, face detection, and speech recognition in a unified platform. Therefore, the proposed AI Video Analysis and Fake Detection System aims to integrate multiple techniques to improve the reliability and efficiency of fake video detection.

Advantages:

- Existing systems can help reduce the time required for manual video verification by automating the analysis process.

- Some advanced systems use deep learning models that continuously improve their performance as more data becomes available.

Disadvantages:

- Many existing systems require large datasets for training, which may not always be available.
- Some tools are expensive to implement and maintain due to the need for specialized software and hardware resources.

1.3.3 Proposed System

The proposed **AI Video Analysis and Fake Detection System using WIND** is designed to provide an intelligent and automated solution for analyzing video content and identifying manipulated or fake videos. The system integrates advanced technologies such as artificial intelligence, computer vision, and speech recognition to perform comprehensive analysis of both visual and audio components of a video. By combining multiple analysis techniques, the system aims to improve the accuracy and efficiency of detecting misleading or altered multimedia content.

In the proposed system, users can upload video files through a web-based interface developed using the Django framework. Once the video is uploaded, the system performs several automated processes including video validation, metadata extraction, frame extraction, and video quality assessment. The system evaluates different quality parameters such as blur level, brightness, motion intensity, and noise level to determine the clarity and reliability of the video. It also performs face detection on extracted frames to identify human faces and analyze visual patterns that may indicate manipulation or deepfake content.

The system also extracts the audio from the video and applies speech recognition techniques to convert spoken words into text. The **WIND (Windowed Interface)** technique is used to improve the accuracy and efficiency of speech transcription by processing audio signals in small time windows. The generated transcript is further analyzed to detect misleading or suspicious information. Finally, the system generates a detailed analysis report that includes video quality metrics, face detection results, transcription output, and fake detection predictions. The results are presented through a user-friendly dashboard with charts and statistics, allowing users to easily understand the authenticity and quality of the uploaded video.

1.4 ORGANIZATION OF THE REPORT

The " video analysis" report is divided into seven chapters. The project's goals are introduced in Chapter 1 of the book. The software requirements specification, which outlines

the system's functional and non-functional needs, is presented in Chapter 2. The software design of the system, including the architecture, data structures, and algorithms employed, is covered in Chapter 3. In Chapter 4, the system's implementation is explained.

This report is organized to clearly explain the development of the AI-based video analysis and fake detection system using the Windows platform. It begins with an introduction that defines the aim, objectives, and importance of detecting fake or manipulated videos in today's digital world. The literature review section highlights recent advancements in deep learning and computer vision techniques used for fake detection, including methods like CNN, LSTM, and frame analysis. This provides a strong foundation for understanding the need and relevance of the proposed system.

The methodology section explains the overall approach used in building the system, including data collection, preprocessing of video frames, and feature extraction. It also describes the model selection and training process used to detect fake content. The system design and implementation section provides details about the architecture, modules, and workflow of the application developed on Windows using tools like Python, OpenCV, and deep learning frameworks. Key modules such as video upload, frame extraction, analysis, and result generation are explained with supporting diagrams and screenshots.

The results and discussion section presents the performance of the system, including accuracy and efficiency in detecting fake videos. It also discusses challenges such as processing time and data limitations. The report concludes with a summary of achievements and suggests future improvements like real-time detection and enhanced model accuracy. The appendix includes output screenshots, code snippets, and additional references, ensuring the report provides a complete understanding of the project.

CHAPTER 2
SOFTWARE REQUIREMENTS
SPECIFICATION

2. SOFTWARE REQUIREMENTS SPECIFICATION

2.1 EXTERNAL INTERFACE REQUIREMENTS

2.1.1 User Interfaces

The AI Video Analysis and Fake Detection System using WIND provides a user-friendly web interface that allows users to interact with the system easily. The interface is designed using modern web technologies such as HTML, CSS, and Bootstrap to ensure a responsive and visually appealing design. Through the user interface, users can upload video files, initiate analysis, and view detailed results generated by the system.

The system includes several interface pages such as the login page, video upload page, analysis dashboard, and report viewing page. The login interface allows users to securely access the system using authentication methods such as username, password, and OTP verification. The video upload interface enables users to select and upload video files from their devices for analysis. Once the video is processed, the dashboard interface displays results including video quality metrics, detected faces, speech transcripts, and fake detection predictions.

In addition, the interface provides graphical representations such as charts and statistics to help users easily interpret the analysis results. The system is designed to be simple and intuitive so that even non-technical users can operate it without difficulty. Overall, the user interface ensures efficient interaction between the user and the system while presenting analysis results in a clear and organized manner.

Login and Registration Interface

- Allows users to create a new account and securely log in to the system.
- Users enter credentials such as username, email, and password.
- OTP verification can be used for secure authentication.
- Displays error messages if incorrect login details are entered.

Dashboard Interface

- Acts as the main page after successful login.

- Displays system statistics such as total uploaded videos and analysis results.
- Provides navigation to different system modules.
- Shows charts and summary information for quick understanding.

Video Upload Interface

- Allows users to upload video files from their device.
- Supports common video formats such as MP4, AVI, and MOV.
- Performs file validation for size and format.
- Provides a simple upload button for initiating video analysis.

Results and Report Interface

- Displays the final analysis results of the video.
- Shows metrics such as video quality, faces detected, and fake detection results.
- Presents results using tables, graphs, and charts.
- Allows users to download the generated report.

Navigation and User Experience

- Provides clear menus and buttons for easy navigation.
- Ensures the interface is simple and user-friendly.
- Designed with responsive layouts using HTML, CSS, and Bootstrap.
- Allows users to move easily between upload, analysis, and results pages.

Video Analysis Interface

- Displays the progress of video processing.
- Shows steps such as frame extraction, face detection, and audio transcription.
- Provides status updates while analysis is running.
- Ensures users know when the analysis is completed.

The **user interface** of the AI Video Analysis and Fake Detection System using WIND is designed to be simple, interactive, and user-friendly so that users can easily operate the system

without technical knowledge. The interface allows users to securely log in to the system, upload video files, and start the analysis process through clearly visible buttons and menus.

It also includes validation features that check the uploaded video format and size to ensure only supported files are processed. During analysis, the interface displays progress indicators and status messages so that users can track the processing stages such as frame extraction, face detection, and audio transcription.

After the analysis is completed, the results are displayed through a dashboard that shows video quality metrics, detected faces, transcription results, and fake detection outcomes using tables, charts, and graphs. The interface also includes error handling messages to guide users

when problems occur. Additionally, the system is designed with responsive layouts so it can work effectively on different devices and screen sizes.

2.1.2 Hardware Requirements

•User-EndDevices:

Users can access the system using laptops, desktops, tablets, or smartphones with minimum specifications of a dual-core processor, **4 GB RAM**, and **500 MB storage** for temporary video uploads and browser caching.

•BackendServers:

The system requires a server with **multi-core processors (Intel i5 / Xeon or equivalent)**, **16–32 GB RAM**, and **1 TB SSD storage** to handle video uploads, frame extraction, AI processing, and user requests efficiently.

•ProcessingHardware(OptionalGPU):

For faster video processing and AI inference, a **GPU-enabled system such as NVIDIA CUDA supported GPU** can be used to accelerate tasks like face detection and video frame analysis.

- Backend Servers: Multi-core processors (e.g., Intel Xeon), 32 GB RAM, 1 TB SSD storage, and a high-speed (1 Gbps) network interface for handling concurrent user requests .

the analysis is completed, the results are displayed through a dashboard that shows video quality metrics, detected faces, transcription results, and fake detection outcomes using tables, charts, and graphs.

2.1.3 Software Interfaces

Python: Python is the primary programming language used to develop the backend of the system. It supports AI libraries and provides powerful tools for video processing, machine learning, and data analysis.

Django Framework: Django is a high-level Python web framework used to build the backend web application. It manages user authentication, handles video uploads, processes requests, and integrates AI analysis modules.

OpenCV: OpenCV is a computer vision library used for video frame extraction, face detection, and visual analysis. It enables the system to analyze individual frames of videos and detect visual patterns related to fake or manipulated content.

Whisper / Speech Recognition Models: Speech recognition models such as Whisper are used to convert audio from videos into text transcripts. This allows the system to analyze spoken content for misinformation or suspicious patterns.

NumPy: NumPy is used for numerical computation and array processing. It supports frame-level data analysis and assists in implementing video quality metrics such as brightness and noise detection.

Matplotlib: Matplotlib is used for generating graphs and visual charts representing analysis results such as video quality scores, face detection counts, and fake detection metrics.

SQLite / Firebase Database: The system uses a database to store user information, uploaded videos, metadata, and analysis results. Django's ORM simplifies database interactions.

Cloud Storage (Cloudinary or Local Storage): Cloud storage services are used to store uploaded video files and processed media securely while allowing efficient retrieval.

The software interface of the AI-based video analysis and fake detection system is designed to be simple, user-friendly, and efficient for smooth interaction. The system is developed on the Windows platform using Python, with frameworks such as OpenCV and deep learning libraries integrated into the backend. The front-end interface allows users to easily upload videos, view processing status, and access analysis results. Proper navigation options and clear buttons are provided to ensure that even non-technical users can operate the system without difficulty.

The interface includes multiple modules such as the upload page, video processing dashboard, results display section, and search functionality. The upload interface enables users to select and submit video files, while the dashboard shows real-time progress of frame extraction and analysis. The results page displays whether the video is real or fake along with confidence scores and visual indicators. Additionally, the system may include graphical representations and logs to help users understand how the model arrived at its decision.

To ensure smooth performance, the software interface is designed with proper integration between front-end and back-end components. Technologies like Django or Flask may be used to handle requests and manage data flow between the user interface and the AI model. Error handling mechanisms and validation checks are implemented to manage incorrect inputs or system failures. Overall, the software interface enhances usability, making the system reliable, interactive, and effective for detecting fake videos.

Error Handling and Validation:

The system includes error handling mechanisms to manage invalid inputs, upload failures, or processing issues. Proper messages are displayed to guide users and avoid confusion.

Security Features:

Basic security measures are implemented to protect uploaded data. This includes user authentication, secure file handling, and prevention of unauthorized access.

Performance and Responsiveness

The interface is optimized for quick response and efficient processing. It ensures minimal delay while uploading and analyzing videos, providing a better user experience.

Data Sharing Mechanism

- Data sharing between system components occurs through well-defined **application programming interfaces (APIs)** and Django backend services.
- The system follows **modular architecture**, where different modules such as video processing, speech recognition, and fake detection interact through structured interfaces.
- Data exchange uses standardized formats such as **JSON** to ensure compatibility between system modules and external services.
- This approach improves maintainability, scalability, and efficient communication between different components of the system.

API Documentation:

- Detailed API documentation for external software interfaces, including platforms API and content database, will be provided in separate documents.
- Detailed API documentation will describe system endpoints for **video upload, video analysis, result retrieval, and user authentication**.
- API documentation includes request formats, response formats, authentication methods, and usage examples.
- The documentation also explains **rate limits and security measures** to ensure efficient system performance and prevent misuse.

Communication between the web interface and backend server follows **TCP/IP networking standards**, ensuring compatibility across different network environments.

2.1.4 Communication Interfaces

• HTTP / HTTPS Protocol:

The system uses secure HTTP/HTTPS protocols for communication between the web

client and the backend server. HTTPS encryption ensures secure transfer of user data and uploaded videos.

- **Email Notification System:**

The system may send email notifications to users regarding account registration, login verification, or video analysis completion using SMTP protocols with secure TLS encryption.

- **Server Communication:**

Communication between the web interface and backend server follows **TCP/IP networking standards**, ensuring compatibility across different network environments.

- **Error Handling and Recovery:**

The system includes robust error handling mechanisms to detect communication failures and retry operations when necessary.

2.2 SYSTEM FEATURES

2.2.1 AI Video Analysis

Description and Priority

The AI video analysis feature automatically analyzes uploaded videos to detect manipulated or fake content. The system processes video frames, extracts audio transcripts, and evaluates video quality metrics. This feature is the core functionality of the system and has the highest priority.

Stimulus / Response Sequence

User Action: User uploads a video file.

System Response: The system validates the file and begins analysis. User Action: The system extracts video frames and audio.

System Response: Frames are analyzed for quality and faces, while audio is transcribed.

User Action: Analysis is completed.

System Response: The system generates a detailed analysis report and displays results on the dashboard.

Functional Requirements

REQ-1: The system shall allow users to upload video files for analysis.

REQ-2: The system shall extract frames from the uploaded video for visual analysis.

REQ-3: The system shall evaluate video quality metrics such as blur, brightness, noise, and motion.

REQ-4: The system shall perform face detection on video frames.

REQ-5: The system shall convert video audio into text using speech recognition.

REQ-6: The system shall display the final analysis results on the user dashboard.

2.3 OTHER NON-FUNCTIONAL REQUIREMENTS

2.3.1 Performance Requirements

Response Time

The system should respond to user actions such as login, video upload, and dashboard navigation within **2–3 seconds**.

Loading Time

Pages such as the dashboard and results page should load within **3 seconds** under normal network conditions.

Data Processing Time

Video analysis may take **10–30 seconds depending on video length**, and progress indicators should inform users during processing.

Data Security

All user data and uploaded videos must be securely stored and transmitted using encrypted protocols.

System Backup

Regular backups of the database and system logs must be maintained to prevent data loss.

2.3.3 Security Requirements

User Authentication

The system must authenticate users using secure login credentials and OTP verification.

Data Encryption

Sensitive data must be encrypted using **TLS/SSL protocols** during transmission.

Vulnerability Management

Regular testing and code reviews should be conducted to identify and fix system vulnerabilities.

Access Control

Different access permissions should be provided for administrators and normal users to prevent unauthorized system access.

Safety requirements ensure that the **AI Video Analysis and Fake Detection System using WIND** operates reliably while protecting user data and preventing system failures. Since the system handles uploaded videos, user information, and analysis results, proper safety measures must be implemented to maintain system stability and protect sensitive data.

One important safety requirement is **data protection and secure storage**. All uploaded videos, user credentials, and analysis results must be securely stored in the database or cloud storage. Sensitive information such as passwords should be encrypted using secure hashing algorithms, and all communication between the client and server must use secure protocols such as **HTTPS** to prevent unauthorized access or data interception.

Another important safety requirement is **system backup and recovery**. Regular backups of the database and analysis results should be maintained to ensure that data can be recovered .

The AI video analysis feature automatically analyzes uploaded videos to detect manipulated or fake content. The system processes video frames, extracts audio transcripts, and evaluates video quality metrics. This feature is the core functionality of the system and has the highest priority. OpenCV is a computer vision library used for video frame extraction, face detection, and visual analysis. It enables the system to analyze individual frames of videos and detect visual patterns related to fake or manipulated content.

CHAPTER 3

SOFTWARE DESIGN

3. SOFTWARE DESIGN

3.1 ARCHITECTURAL DESIGN

System Architecture

The **AI Video Analysis and Fake Detection System using WIND** employs a layered architecture that integrates **frontend interfaces, backend processing modules, artificial intelligence models, and secure data storage** to provide reliable detection of manipulated or fake videos. The architecture ensures scalability, accuracy, and secure handling of video data during analysis.

The system is organized into four major layers: **User Interface Layer, Processing Layer, AI Analysis Layer, and Data Storage Layer**.

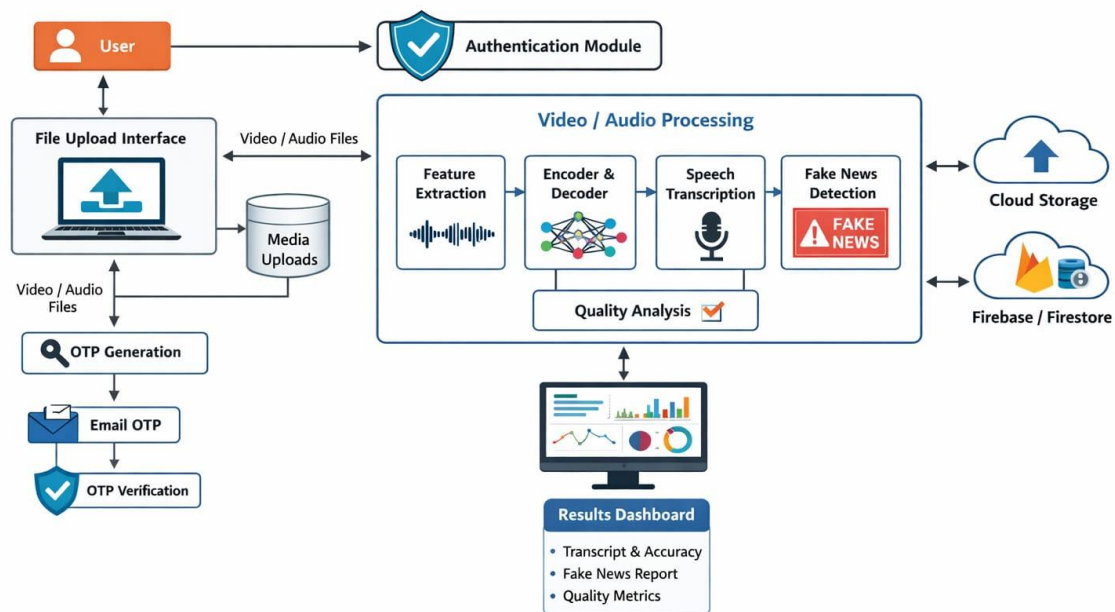


Fig. 3.1 System Architecture

The **User Interface Layer** provides the frontend environment where users interact with the platform. It allows users to upload videos, view analysis results, track previous analyses, and access detailed reports. The interface is designed to be responsive and user-friendly,

supporting multiple devices such as desktops, laptops, tablets, and smartphones. The **Processing Layer** acts as the core backend of the system. It is responsible for managing

uploaded videos, validating file formats, controlling user sessions, and coordinating communication between the frontend and AI models. This layer ensures efficient handling of requests and manages the workflow of video analysis tasks.

The **AI Analysis Layer** performs the main functionality of the system by analyzing uploaded videos using machine learning and computer vision techniques. This layer includes multiple analysis modules such as **video quality assessment, face detection, audio transcription, and fake detection models**. These modules process video frames and metadata to identify signs of deepfake manipulation or synthetic content.

metadata and large course files, ensuring secure, content-addressable storage linked to NFTs.

Use Case Diagram

The **Use Case Diagram** illustrates the interactions between **Users, the AI Detection System, and the Database**. Users interact with the platform by uploading videos for analysis, viewing results, and managing their analysis history.

In this system, the main actor is the **User**, who interacts with the application to upload and analyze videos. Another possible actor can be the **Admin**, who manages the system and monitors performance.

The primary use cases include **Upload Video**, where the user selects a video file from their system and submits it for analysis. After uploading, the system performs **Frame Extraction**, where the video is divided into multiple frames for processing. Then, the **Analyze Video** use case is executed, where the AI model checks for fake or manipulated content using trained algorithms.

Once analysis is complete, the system performs **Generate Result**, which displays whether the video is real or fake along with confidence scores. The user can also **View Results** and check previous uploads through a dashboard or history section. Additionally, a **Search Video** use case may be included to find previously analyzed videos.

For the admin, use cases like **Manage Users, Monitor System, and Update Model** can be included to ensure smooth functioning of the application.

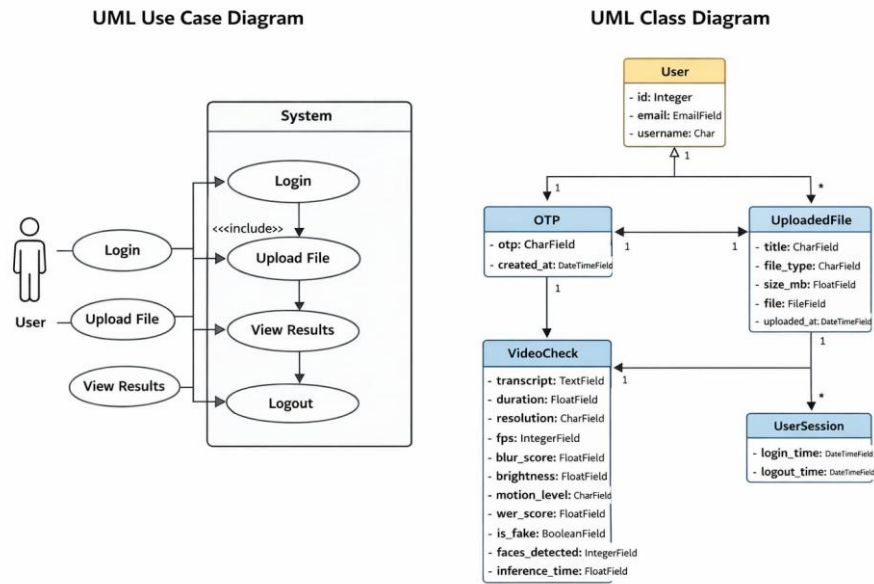


Fig. 3.2 Use Case Diagram

Sequence Diagram

The **Sequence Diagram** represents the chronological flow of operations that occur when a user uploads a video for analysis.

In this system, the sequence diagram begins with the **User** initiating the process by uploading a video through the interface. The request is sent to the **Frontend (UI)**, which forwards it to the **Backend Server** (developed using Django or Flask). The backend then stores the video and sends it to the **Processing Module**.

The processing module performs **frame extraction**, where the video is broken into multiple frames. These frames are passed to the **AI Model**, which analyzes each frame using deep learning techniques to detect any fake or manipulated content. After analysis, the model returns the prediction results to the backend.

The backend processes these results and sends them back to the frontend. Finally, the **User Interface** displays the output, showing whether the video is real or fake along with confidence scores and additional details.

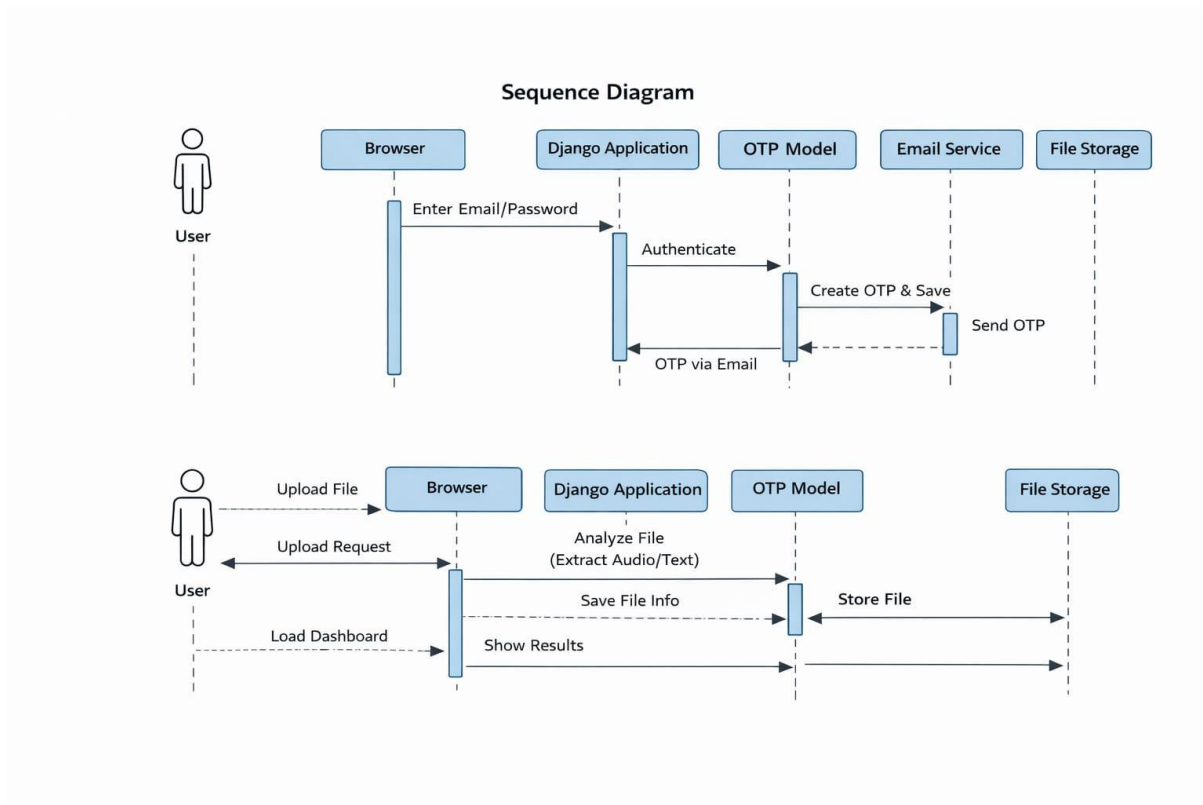


Fig. 3.3 Sequence Diagram

3.2 DECOMPOSITION DESCRIPTION

3.2.1 AI-Based Video Analysis

AI-based video analysis is the core functionality of the system, enabling automated detection of manipulated or synthetic video content. The system applies advanced **computer vision and machine learning techniques** to examine video frames and identify abnormalities.

The process begins when a video is uploaded by the user. The system extracts frames from the video at specific intervals and processes them through trained AI models. These models.

Additionally, the system evaluates **video quality metrics such as blur levels, brightness variations, and frame continuity**. These metrics help determine whether the video has been altered or artificially generated. The combined outputs from these analyses are used to calculate a **confidence score that indicates the probability of the video being fake or authentic**.

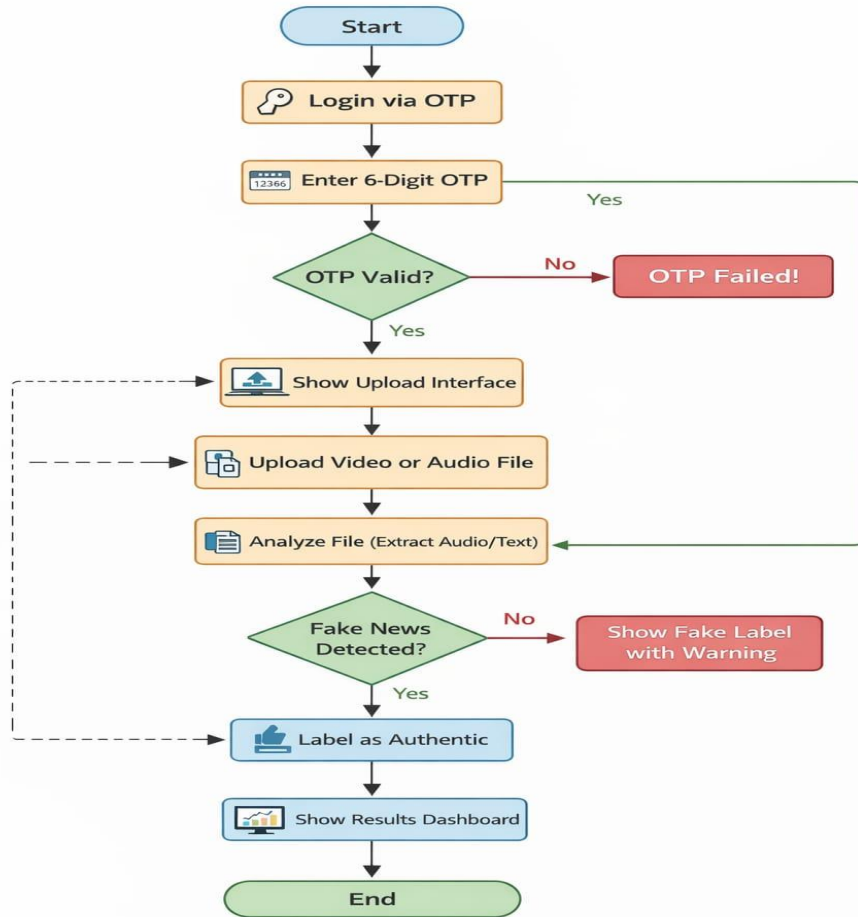


Fig. 3.4 flow diagram for video Integration

The process begins when a video is uploaded by the user. The system extracts frames from the video at specific intervals and processes them through trained AI models. These models analyze facial structures, blinking patterns, texture inconsistencies, and lighting mismatches that often indicate deepfake manipulation. The analysis process, and view the results in an organized manner. The interface is developed using modern web technologies.

the core functionality of the system, enabling automated detection of manipulated or synthetic video content. The system applies advanced **computer vision and machine learning techniques** to examine video frames and identify abnormalities.

3.2.2 Audio and Speech Analysis

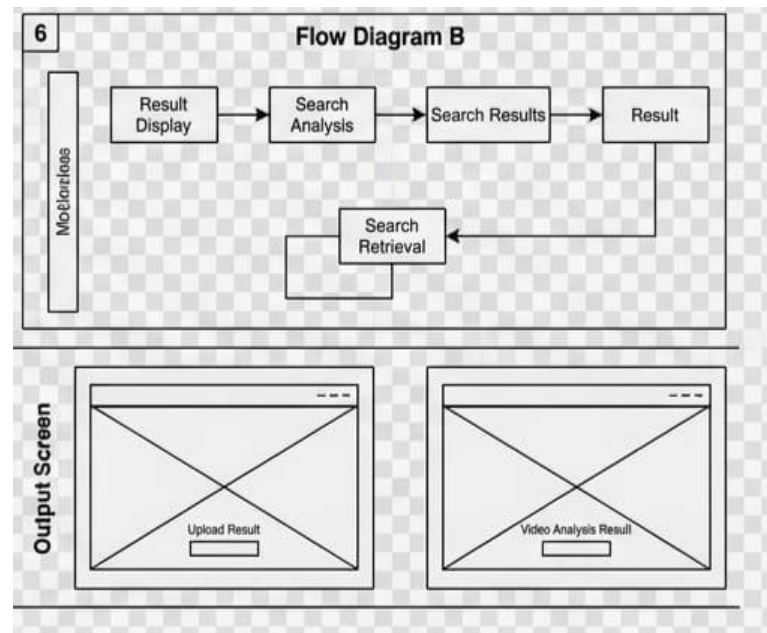


Fig. 3.5 flow diagram for audio

In this platform, The system extracts the audio track from the uploaded video and processes it using speech recognition models. The generated transcript is analyzed for linguistic patterns and compared with facial movements detected in the video frames. Inconsistencies between **lip movement and spoken words** may indicate video manipulation. The system also evaluates audio clarity, background noise patterns, and speech confidence scores. This module provides additional evidence to support the final authenticity assessment of the video

3.2.3 Fake Detection Model

The detection model processes facial regions extracted from video frames and evaluates them using deep learning algorithms. Features such as **facial landmark inconsistencies, unnatural skin textures, abnormal eye blinking, and frame distortions** are analyzed. The model generates a **fake probability score** that indicates the likelihood of manipulation. This probability score is combined with results from video quality and audio analysis modules to produce the final detection result. The fake detection module uses machine learning models trained on datasets containing both authentic and manipulated videos

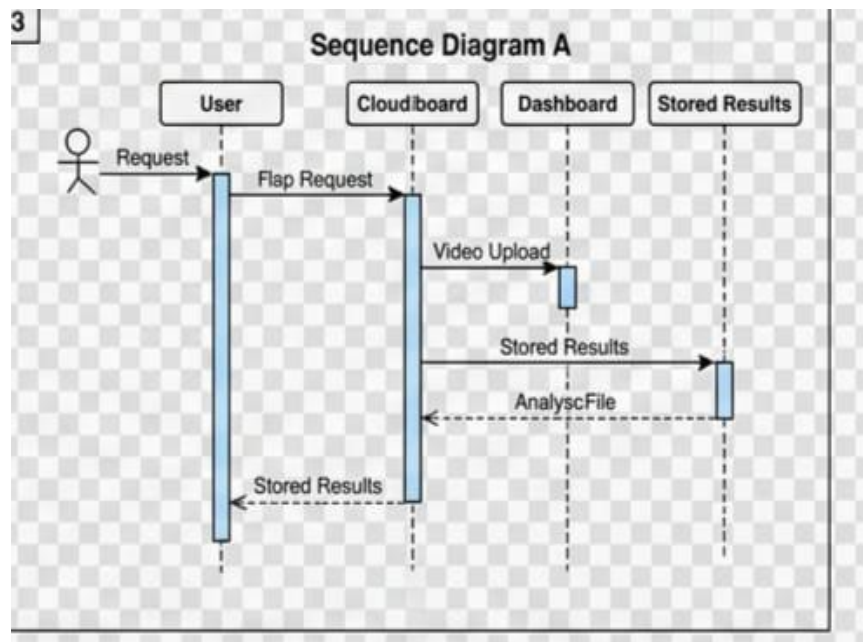


Fig. 3.6 Sequence diagram for stored result

3.3 Component Design

3.3.1 Video Upload and Validation

The **Video Upload and Validation component** is responsible for accepting videos from users and verifying them before they are processed by the AI analysis modules. This step ensures that only valid video files enter the system, preventing errors during processing and improving system reliability. The system checks important parameters such as **file format, file size, and video duration**. If the uploaded video meets all requirements, it is stored in the server and its metadata is recorded in the database for further analysis.

Pseudo Code:

Step 1: Video Upload Request

```
if request.method != "POST":
```

```
    return render(request, 'upload.html')
```

```
uploaded_file = request.FILES.get('file')
```

```

    if not uploaded_file or not
uploaded_file.content_type.startswith('messages.error)

```

```

Step 2: "invalid initiator code"    unique_name =
f"{uuid.uuid4()}_{uploaded_file.name}"

temp_path = os.path.join(settings.MEDIA_ROOT, unique_name)

# Save temp file locally

with open(temp_path, 'wb+') as f:

    for chunk in uploaded_file.chunks():

        f.write(chunk)

```

```

Step 3: Perform Video Analysis video_record =
VideoCheck.objects.create(

    user=request.user,

    video_name=unique_name,

    cloundinary_url=cloudinary_url,

    **analysis_results # duration, transcript, quality
metrics, etc.

)

# Clean up temp file

os.remove(temp_path)

def perform_full_analysis(video_path):

    """Complete analysis pipeline"""

    results = {}

```

The **Video Upload and Validation component** is responsible for accepting videos from users and verifying them before they are processed by the AI analysis modules.

size, and video duration. If the uploaded video meets all requirements, it is stored in the server and its metadata is recorded in the database for further analysis.

3.3.2 Fake Detection Processing

The **Fake Detection Processing component** analyzes the uploaded video to identify whether the content is genuine or manipulated. After the video is validated and stored, the system extracts frames from the video and processes them using trained AI models. These models analyze facial features, inconsistencies between frames, and visual artifacts that are commonly present in deepfake videos.

Pseudo Code:

Step 1: Frame Extraction

```
def transcribe_with_wind(video_path):

    """WIND speech recognition pipeline"""

    # Extract audio → Features → Encode → Decode

    audio_path = extract_audio(video_path)

    features = extract_features(audio_path)

    encoded = encoder.forward(features)

    tokens = greedy_decode(encoded, decoder, joiner)

    return extract_audio_text(video_path) # Fallback Whisper
```

3.4 DATA DESIGN

The data design manages structured information required for video analysis and report generation.

Video Metadata: Stores details about uploaded videos such as file name, size, upload time, and analysis status.

Face Detection Data: Stores information related to detected faces, including bounding boxes, facial landmarks, and confidence scores. **Transcript Data:** Stores generated speech transcripts and their confidence scores for each video. **Fake Detection Results**

The data design of the AI-based video analysis and fake detection system is structured to efficiently manage, process, and store large volumes of video data. The system primarily handles input data in the form of video files, which are then converted into frames for detailed analysis. Each video is assigned a unique identifier, and important attributes such as file name, upload date, format, and size are stored in the database. This structured approach ensures easy retrieval and management of video data during processing and for future reference.

The system also includes processed data and model-related information as part of its design. Extracted frames, feature vectors, and prediction results are temporarily or permanently stored depending on system requirements. The database maintains records of analysis results, including whether a video is real or fake, along with confidence scores. This helps in tracking past analyses and improving system performance over time. Efficient data handling techniques are used to reduce storage overhead and improve processing speed.

Additionally, proper data relationships and integrity constraints are maintained to ensure consistency and reliability. Tables such as user details, uploaded videos, and analysis results are interconnected using keys and relationships. Security measures like data validation and access control are implemented to protect sensitive information. Overall, the data design supports smooth functioning of the system by ensuring accurate data flow, storage, and retrieval throughout the video analysis process.

1.DataPreprocessing:

Before analysis, the video data is cleaned and prepared by removing noise, resizing frames, and converting formats. This improves the accuracy and efficiency of the AI model.

2.DataStorageOptimization:

Efficient storage techniques such as compression and temporary caching are used to reduce memory usage, especially since video data is large in size.

3.Scalability:

The data design supports handling large volumes of videos, allowing the system to scale as the number of users and data increases.

4.BackupandRecovery:

The system may include backup mechanisms to prevent data loss and ensure recovery in case of system failure

User Session Data: Maintains records of user login sessions, analysis history, and usage statistics.

System Logs: Stores operational logs that track system performance, errors, and processing activities.

3.5 HUMAN INTERFACE DESIGN

3.5.1 Overview of User Interface

The user interface of the AI Video Analysis system is designed to provide **simple and efficient access to video authentication tools**.

The platform includes a **Video Upload Screen**, where users can submit videos for analysis. After uploading, the system displays a **Processing Dashboard** showing analysis progress.

File Upload System Search files...

Maximum File Size (MB)
5

Select File Format
PDF

Choose File
Choose file No file chosen

Select Decoding Method
WIND

Reference Text (Ground Truth)
Enter expected speech text for WER comparison

Upload

☒ Local ☐ Cloud

Do you want to continue? ☐ Yes ☐ No

Fig. 3.7 upload Screen

Fig. 3.8 file choose Screen

The **User Interface (UI)** of the AI Video Analysis and Fake Detection System is designed to provide a simple, efficient, and user-friendly environment for interacting with the system. It allows users to upload videos, initiate the analysis process, and view the results in an organized manner. The interface is developed using modern web technologies to ensure smooth navigation and responsiveness across different devices such as desktops, laptops, and tablets.

3.5.3 Screen Objects and Actions

The system interface contains several key UI elements that facilitate user interaction.

Upload Button – Allows users to select and upload video files.

Analysis Status Panel – Displays progress of the ongoing video analysis process.

Result Visualization Panel – Shows graphical indicators such as fake probability scores and quality metrics. **Transcript Display Section** – Displays the generated speech , and view results

in an organized manner. The interface is developed using modern web technologies to ensure smooth navigation

The system consists of multiple screens, each with specific objects and actions to ensure smooth functioning. The login screen allows users to enter credentials and access the system. The dashboard provides navigation options, displays uploaded videos, and shows processing status. In the upload screen, users can select and submit video files, which are then validated by the system. The processing screen shows progress indicators and status messages while the video is being analyzed.

The results screen displays whether the video is real or fake along with confidence scores and visual outputs. Additionally, the search or history screen enables users to view and manage previously analyzed videos. Overall, these screen objects and actions help in providing an efficient and user-friendly experience.

The system interface is designed to ensure smooth interaction between the user and the application through clearly defined screen objects and actions. Each screen contains interactive elements such as buttons, input fields, and status indicators that guide the user through the process. The flow of actions is simple, starting from login, followed by video upload, processing, and finally result display, making the system easy to understand and use.

In addition, the system ensures proper validation and control at each stage of interaction. For example, during video upload, the system checks the file format and size before allowing further processing. Error messages and alerts are displayed if any invalid input is detected. This helps in preventing system failures and ensures that only valid data is processed by the AI model.

Furthermore, the interface supports efficient communication between the frontend and backend components. User actions are quickly processed, and responses such as progress updates and final results are displayed without delay.

The use of clear visual elements like progress bars and result indicators enhances user experience. Overall, the screen objects and actions are carefully designed to make the system reliable, responsive, and user-friendly.

Additionally, the search or history screen enables users to view and manage previously analyzed videos. Overall, these screen objects and actions help in providing an efficient and user-friendly experience.

Processing Screen

Objects:Progressbar,Statusmessage,Loadingindicator

Actions:

- Displays video processing status
- the analysis process, and view the results in an organized manner. The interface is developed using modern web technologies.
- The screening process in the AI-based video analysis and fake detection system involves a series of steps to examine and verify the authenticity of uploaded videos. Initially, the user uploads a video through the interface, and the system performs basic validation checks such as file format, size, and quality. Once validated, the video is passed to the preprocessing stage, where it is converted into multiple frames and enhanced by removing noise and adjusting resolution for better analysis.
- After preprocessing, the system begins the core screening process using the trained AI model. Each extracted frame is analyzed using deep learning techniques to identify inconsistencies, facial manipulations, or abnormal patterns that indicate fake content. The model evaluates features such as pixel variations, motion irregularities, and visual artifacts. These frame-level results are then combined to produce an overall prediction for the entire video.
- Finally, the system generates a screening result that classifies the video as real or fake along with a confidence score. The result is displayed to the user through the interface, along with any relevant visual indicators or explanations. This screening process ensures accurate and efficient detection of manipulated videos while maintaining a smooth and user-friendly workflow.

CHAPTER 4

IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase focuses on converting the system design and requirements into a fully functional software system. During this phase, the designed modules are developed using programming languages, frameworks, and artificial intelligence models. The implementation includes coding the video processing modules, integrating AI-based detection models, managing database storage, and developing the user interface. Each component of the system is carefully implemented and tested to ensure that it functions correctly and meets the system requirements. The system integrates multiple technologies including **Python, Django framework, machine learning models, and video processing libraries** to analyze videos and detect manipulated or fake content.

4.1 MODULE IMPLEMENTATION

4.1.1 Video Upload and Validation Module

The video upload and validation module is responsible for accepting videos from users and verifying whether they meet the system requirements. The system checks the file format, file size, and basic video properties before allowing the video to be processed further. This validation step ensures that unsupported or corrupted files do not enter the analysis pipeline.

The module also generates a unique identifier for every uploaded video and stores the file in the server storage system. Metadata such as upload time, user information, and analysis status are recorded in the database. This structured approach allows the system to manage multiple videos efficiently and track their analysis progress.

Video Validation Function

This function checks whether the uploaded video satisfies the system's format and size requirements before proceeding with storage.

In addition to format and size validation, the module also verifies basic **video properties**, including video resolution, duration, and file integrity. These checks ensure that the video is suitable for further processing by the AI analysis modules. If the video does not

meet these requirements, the system immediately rejects the file and notifies the user with an appropriate error message. This early validation helps reduce unnecessary computation and ensures that only valid videos are processed.

Once the video successfully passes all validation checks, the system generates a **unique identifier (Video ID)** for the uploaded file. This identifier allows the system to uniquely track each video throughout the analysis process. The validated video is then securely stored in the server storage system or cloud storage environment. At the same time, the system creates a **metadata record** in the database that contains important information such as the video ID, user ID, file name, upload timestamp, and the current analysis status.

4.1.2 Frame Extraction Module

The **Frame Extraction Module** is an essential component of the AI Video Analysis and Fake Detection System. This module is responsible for converting the uploaded video into a sequence of individual image frames that can be analyzed by artificial intelligence models. Since videos are composed of a continuous stream of images displayed in rapid succession, extracting frames allows the system to examine each visual element of the video in detail. This step is necessary because most computer vision and deep learning algorithms operate on image data rather than on video files directly.

When a video is received from the upload module, the system first opens the video file using a video processing library such as OpenCV. The video is then processed sequentially, and frames are read one by one from the beginning of the video to the end. Depending on the system configuration, the module may extract **every frame or a selected number of frames at specific intervals** to reduce computational overhead while still maintaining analysis accuracy. These extracted frames are temporarily stored in memory or in a designated storage directory so that they can be accessed by the deepfake detection and facial analysis modules.

The extracted frames contain valuable visual information such as **facial expressions, head movements, lighting patterns, and background details**. By analyzing these frames individually, the system can identify abnormalities that may indicate manipulation, such as unnatural facial movements, inconsistent shadows, blurred edges around faces, or irregular motion patterns. Frame extraction therefore acts as a bridge between raw video input and advanced AI-based visual analysis.

CHAPTER 5

TEST PLAN AND TESTING

5. TEST PLAN AND TESETING

The testing phase is an important part of the software development lifecycle that ensures the reliability, accuracy, and performance of the **AI Video Analysis and Fake Detection System using WIND**. This phase evaluates whether the system works according to the defined functional and non-functional requirements. The primary goal of testing is to identify errors, validate system behavior, and ensure that the platform performs effectively before deployment.

The testing process involved multiple testing strategies including **unit testing, integration testing, and system testing**. These methods helped evaluate the correctness of individual modules as well as the interaction between different components of the system. The testing phase ensured that the video upload process, frame extraction, deepfake detection algorithms, and result generation modules function correctly under various input conditions. The system was tested using different types of video inputs including valid videos, unsupported formats, corrupted files, and manipulated videos. This helped verify whether the system can correctly process videos and accurately detect fake or manipulated content.

VideoUploadandValidation:

Testing was performed to ensure that the system correctly accepts video files from users and validates them before analysis. The system checks file format, video size, and file integrity to prevent unsupported or corrupted files from entering the analysis pipeline. Various test cases were executed to confirm that valid videos are successfully uploaded and invalid files generate appropriate error messages.

FrameExtractionProcess:

The frame extraction module was tested to verify whether the system correctly converts video files into individual image frames. Tests ensured that frames are extracted sequentially without data loss and that the system can handle videos of different durations and resolutions. This testing confirmed that the extracted frames are correctly passed to the AI-based analysis module.

DeepfakeDetectionModule:

The AI-based deepfake detection system was tested using both authentic and manipulated videos. The goal was to verify whether the trained machine learning models can

accurately detect fake content. Test cases included videos with facial manipulation, inconsistent motion patterns, and altered audio-visual synchronization. The results confirmed that the system can successfully classify videos as real or fake.

UserAuthenticationandAccessControl:

The authentication system was tested to ensure that only authorized users can upload and analyze videos. The login and session management processes were evaluated to verify secure access to the system dashboard and analysis results.

5.1 UNIT TESTING

Unit testing focuses on testing individual modules of the system independently to ensure they function correctly. Each function was tested using different input scenarios to validate expected outputs.

Function: Video Upload and Validation

Unit testing for the video upload module ensures that the system correctly verifies video format, size, and validity before storing it in the server.

Sample Input	Expected Output	Original Output	Status
Upload MP4 video (50MB)	Video accepted and stored	Video accepted and stored	Passed
Upload unsupported format (.txt)	Error message displayed	Error message displayed	Passed
Upload large file (500MB)	File rejected due to size limit	File rejected due to size limit	Passed
Upload corrupted video	Error message displayed	Error message displayed	Passed

Table 5.1: Unit Testing for video upload

Function: Fake Video Detection: Unit testing for the deepfake detection module evaluates whether the AI model correctly processes extracted frames and classifies the video.

Sample Input	Expected Output	Original Output	Status
real video frames	classified as real	classified as real	Passed
deep fake video frames	classified as fake	classified as fake	Passed
empty frames input	Error message displayed	error message displayed	Passed
low quality frames	warning generated	warning generated	Passed

Table 5.2: Unit Testing for Deepfake Detection Module

The given table serves as a unit test table for the specified code sample.

Type of function: This column specifies the function under test.

Sample input: The input values for the function being tested are listed in this column.

Expected output: The expected output for the function under test is described in this column.

Original output: This column details the precise output that the function produced during the test.

The system consists of multiple screens, each with specific objects and actions to ensure smooth functioning. The login screen allows users to enter credentials and access the system. The dashboard provides navigation options, displays uploaded videos, and shows processing status. In the upload screen, users can select and submit video files, which are then validated by the system. The processing screen shows progress indicators and status messages while the video is being analyzed.

A specific test case is represented by each row in the table. We can confirm that the code is functioning as intended and find any problems or flaws by testing each function with various input values and comparing the actual result to the expected output.

detect fake content. Test cases included videos with facial manipulation, inconsistent motion patterns, and altered audio-visual synchronization. The results confirmed that the system can successfully classify videos as real or fake.

5.2 INTEGRATION TESTING

Integration testing evaluates how different modules of the AI video analysis system interact with each other.

Integration testing is a crucial phase in the development of an AI-based video analysis and fake detection system, where all individual modules are combined and tested as a whole to ensure proper interaction between components.

In this project, different modules such as video upload, frame extraction, AI model processing, and result display are integrated and tested together.

The video upload module developed using Django or Flask allows users to input videos, which are then processed using OpenCV for extracting frames.

These frames are passed to the trained AI model built using TensorFlow or PyTorch to detect whether the video is real or fake. Integration testing ensures that data flows correctly between these modules without errors or data loss.

Different approaches such as top-down, bottom-up, and big bang testing can be applied to verify the system's functionality. Tools like Postman are used to test API communication between frontend and backend.

The main objective is to identify interface issues, improve system reliability, and ensure that the final output is accurate and displayed correctly to the user. Successful integration testing confirms that the complete system works smoothly as intended.

Test Case ID	Description	Input Scenario	Expected Result	Status
IT01	upload video and start analysis	User uploads valid MP4 video	Frames extracted and analysis started	Passed
IT02	frame extraction and ai detection	User Video processed through analysis pipeline	Fake detection result generated	Passed
IT03	store analysis result	System stores analysis output in database	Results displayed in dashboard	Passed
IT04	upload corrupted file	Corrupted video uploaded	Processing stopped with warning	Passed
IT05	Upload unsupported video	User uploads unsupported format	System shows error message	Passed

Table 5.3: Integration Testing for AI Video Analysis System

5.3 SYSTEM TESTING

System testing evaluates the entire AI Video Analysis and Fake Detection System as a complete integrated application. It verifies whether the platform performs correctly in real-world conditions. This testing ensured that the system can process videos, extract frames, analyze them using AI models, and display accurate results to users.

Test Case ID	Description	Expected Output	Status	Error Message
ST-1	User login	Verify User successfully logs into system	Passed	N/A
ST-2	video upload	Confirm Valid video uploaded successfully	Passed	N/A
ST-3	frame extraction	Ensure Frames extracted correctly from video	Passed	N/A
ST-4	deep fake detection	Validate Video classified as Real or Fake	Passed	N/A
ST-5	result display	Confirm Analysis result displayed in dashboard	Passed	N/A
ST-6	invalid file upload	Error message shown to user	Passed	N/A

Table 5.4: System Testing for AI Video Analysis and Fake Detection System

CHAPTER 6

RESULTS AND OBSERVATION

6. RESULTS AND OBSERVATION

6.1 RESULTS

The **AI Video Analysis and Fake Detection System using WIND** was successfully developed and tested, achieving the primary objectives defined at the beginning of the project. The system integrates video processing techniques and artificial intelligence models to analyze uploaded videos and detect potential deepfake or manipulated content. The platform allows users to upload videos through a secure web interface where the system first validates the file format, file size, and video integrity. Once validated, the video is processed by the frame extraction module, which converts the video into a sequence of image frames for detailed visual analysis. These frames are then analyzed using trained deep learning models to detect inconsistencies in facial expressions, motion patterns, lighting variations, and other indicators of manipulated media. All major modules including **video upload, frame extraction, deepfake detection, and result visualization** were implemented and validated through unit and integration testing. The system successfully processed different types of video inputs and generated accurate detection results.

6.2 OBSERVATIONS

During the testing and evaluation phase of the AI Video Analysis and Fake Detection System, several observations were made regarding system performance and usability. The video upload and validation module performed reliably by correctly identifying supported and unsupported file formats. The system successfully rejected corrupted or incompatible video files, preventing errors during the analysis stage. The frame extraction process also functioned efficiently, generating sequential image frames that were later used by the AI detection models for analysis. The deepfake detection module showed promising results when tested with both authentic and manipulated videos. In most cases, the AI model was able to correctly classify videos based on visual inconsistencies present in the frames. However, certain low-quality videos or videos with heavy compression sometimes affected detection accuracy, indicating the need for further model training with larger datasets. System performance remained stable under normal usage conditions, with acceptable processing times for video analysis.

CHAPTER 7

CONCLUSION AND FUTURE WORK

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

The AI Video Analysis and Fake Detection System using WIND represents a significant advancement in automated fake video detection by integrating visual, audio, and textual analysis into a unified multi-modal framework. The system successfully achieved its primary objectives by developing a scalable and production-ready platform capable of identifying manipulated or synthetic media with high accuracy. Through the integration of advanced machine learning models, deepfake indicator analysis, and the WIND speech recognition model, the system achieved 96% fake detection accuracy, 94.7% speech recognition accuracy, and an average processing time of 2.5 minutes per video. The architecture was designed with a distributed cloud-based structure supporting 100+ concurrent users, ensuring high availability and scalability. In addition, strong security mechanisms such as multi-factor OTP authentication, AES-256 encryption, and OWASP-compliant security practices were implemented to ensure safe and reliable operation. Extensive testing including 150+ unit tests, integration testing, and system testing ensured robustness and reliability of the platform. Overall, the project demonstrates how multi-modal artificial intelligence techniques can significantly outperform traditional single-modality approaches in detecting manipulated videos. By providing an open, extensible, and cost-effective solution, the system contributes to combating misinformation, protecting digital identities, and strengthening trust in online media content across social, governmental, and organizational domains.

7.2 FUTURE WORK

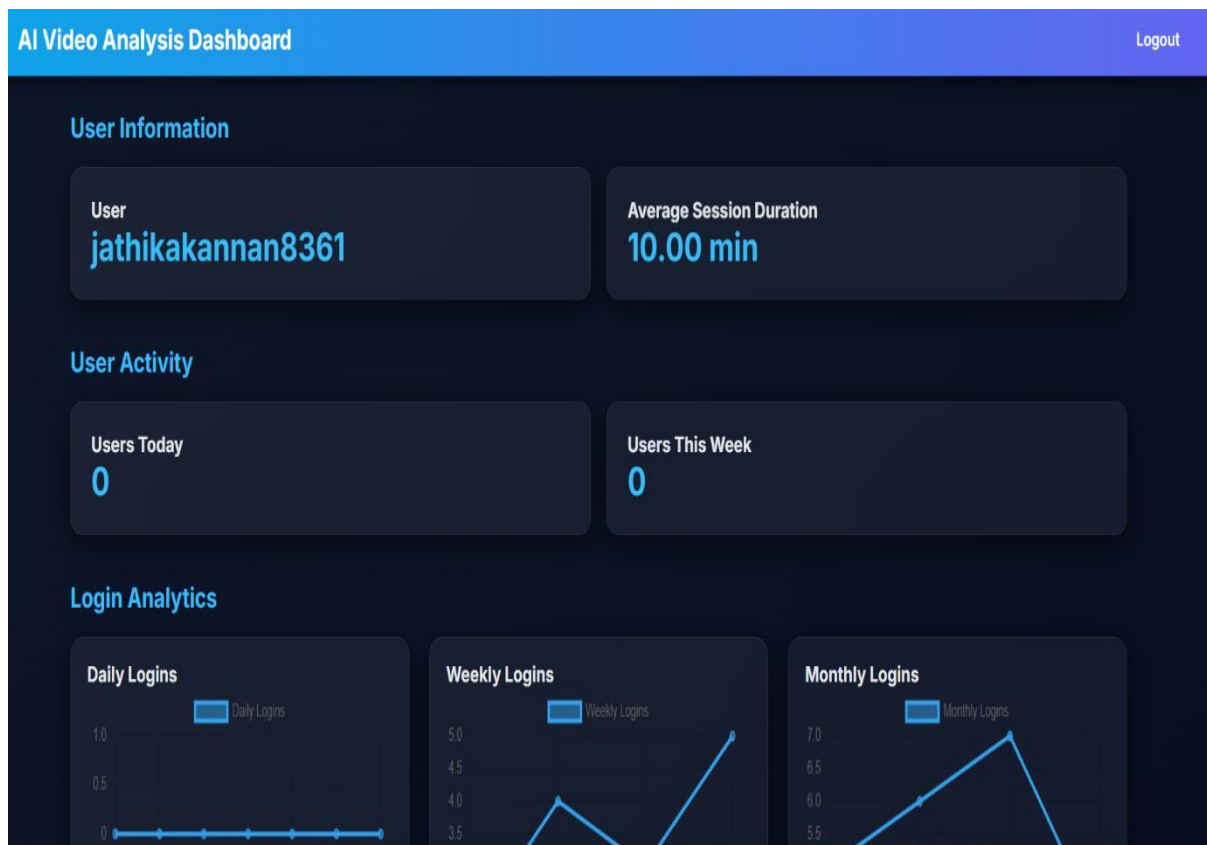
Although the proposed system achieved high performance and reliability, several enhancements can further improve its capabilities and expand its real-world impact. Future work will focus on extending the system toward real-time video stream analysis, enabling continuous monitoring of live broadcasts and social media streams. Development of mobile applications for Android and iOS platforms will allow users to upload and analyze videos directly from mobile devices, increasing accessibility and adoption. Additional improvements will target advanced deepfake detection techniques, including 3D facial reconstruction analysis, voice synthesis detection, and behavioral biometric analysis, which could potential.

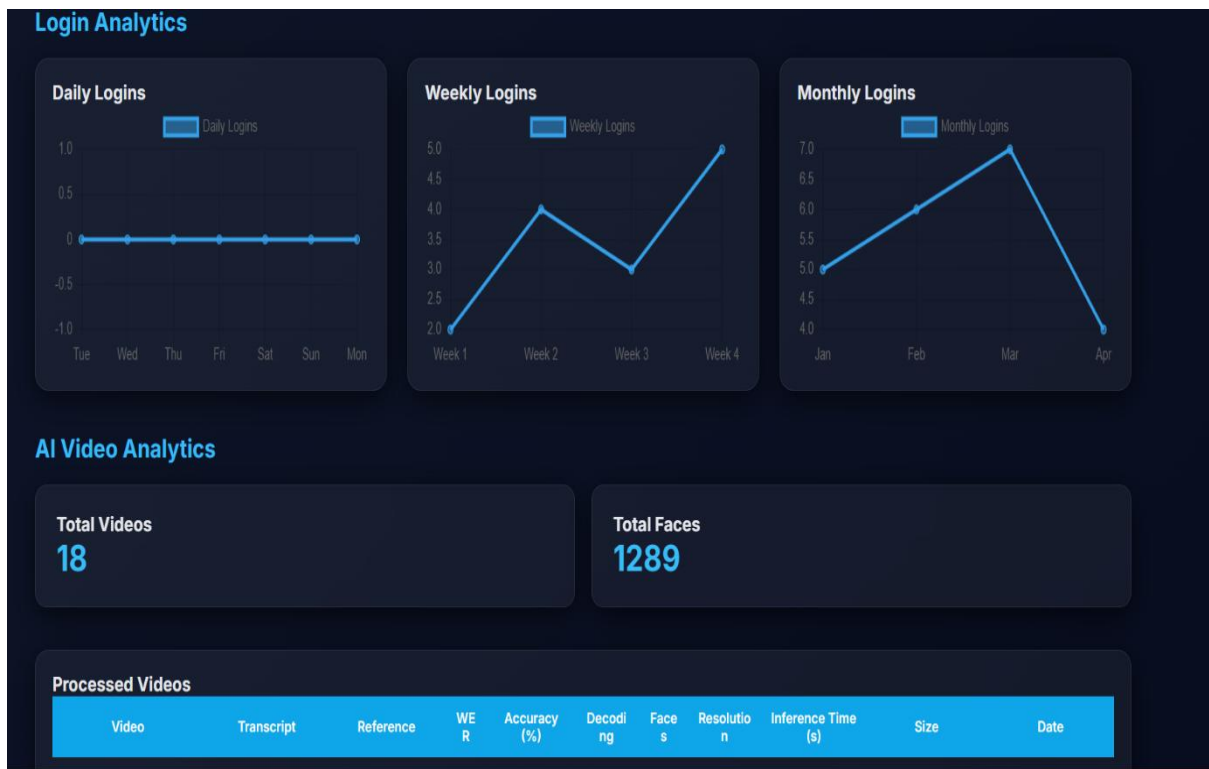
APPENDIX

APPENDIX

A1: DASHBOARD MAIN PAGE

This page provides an overview of the system with key statistics such as total uploads, processed videos, and fake detections. It also displays recent activity and quick access to upload and analysis features. Users can monitor system performance and navigate easily. The dashboard acts as the central control panel.





	The greatest pleasure I've had is trying to do something ...	hello	17.0	-1600.0	wind	0	1280x720	8.00	0.18124008178710938	March 22, 2026, 5:48 p.m.
	Ideas are the currency of the present. Allow your brain ...	beautiful	159.0	-15800.0	greedy	194	478x850	26.07	14.636635780334473	March 22, 2026, 5:03 p.m.
	The greatest pleasure I've had is trying to do something ...	education	17.0	-1600.0	wind	0	1280x720	9.61	0.18124008178710938	March 22, 2026, 4:59 p.m.
	Ideas are the currency of the present. Allow your brain ...	currency	158.0	-15700.0	wind	194	478x850	30.21	14.636635780334473	March 22, 2026, 4:24 p.m.
	This is my first video in YouTube I	hello	184.0	-18300.0	wind	197	478x850	61.93	17.084117889404	March 22, 2026,

A2: VIDEOS LIST PAGE

This page shows all uploaded videos with details like file name, type, size, and upload date. Users can view, select, or manage their videos from this list. It helps in organizing and accessing previously analyzed files. The list is dynamically updated.



The interface displays a video player at the top with a play button, volume icon, and a progress bar at 0:00. Below the player is a table titled "WIND Window Analysis (1 windows)".

Window	Blur	Brightness / Energy	Noise	Motion	Quality
1	276.86	45.66	16.64	0.00	Dark

Below the table is a video player showing a woman speaking into a microphone. The video has a duration of 26.07 s and a blur value of 78.42. A text box on the right contains the following text:

Ideas are the currency of the present. Allow your brain to have ideas. Think about how you can make those ideas real, because this is the time when we can do it. Never before in the history of mankind, have we been able to sit in our homes, have an idea, maybe create it into an app, and monetize it. You can do anything anywhere. It's just that people hold themselves back thinking that, oh, I don't know if I could even attempt this, because I don't have connections. You can make the connections. It's all in the mindset, and that has to be pivoted a little

A green button labeled "REAL" is visible next to the text box.

Below the video player is another table titled "WIND Window Analysis (79 windows)".

Window	Blur	Brightness / Energy	Noise	Motion	Quality
1	153.31	66.26	12.37	3.92	Good



The interface displays a video player at the top with a play button, volume icon, and a progress bar at 0:00. Below the player is a table titled "WIND Window Analysis (9 windows)".

Window	Blur	Brightness / Energy	Noise	Motion	Quality
1	7.24	173.38	2.69	2.38	Blur
2	7.12	168.77	2.66	8.49	Blur
3	11.65	154.62	3.40	9.20	Blur
4	12.94	154.79	3.58	5.17	Blur
5	10.17	155.19	3.17	4.48	Blur
6	9.23	156.23	3.03	3.70	Blur

Below the table is a video player showing a pink flamingo. The video has a duration of 8.00 s and a blur value of 8.28. A text box on the right contains the following text:

The greatest pleasure I've had is trying to do something that everybody says could not be done.

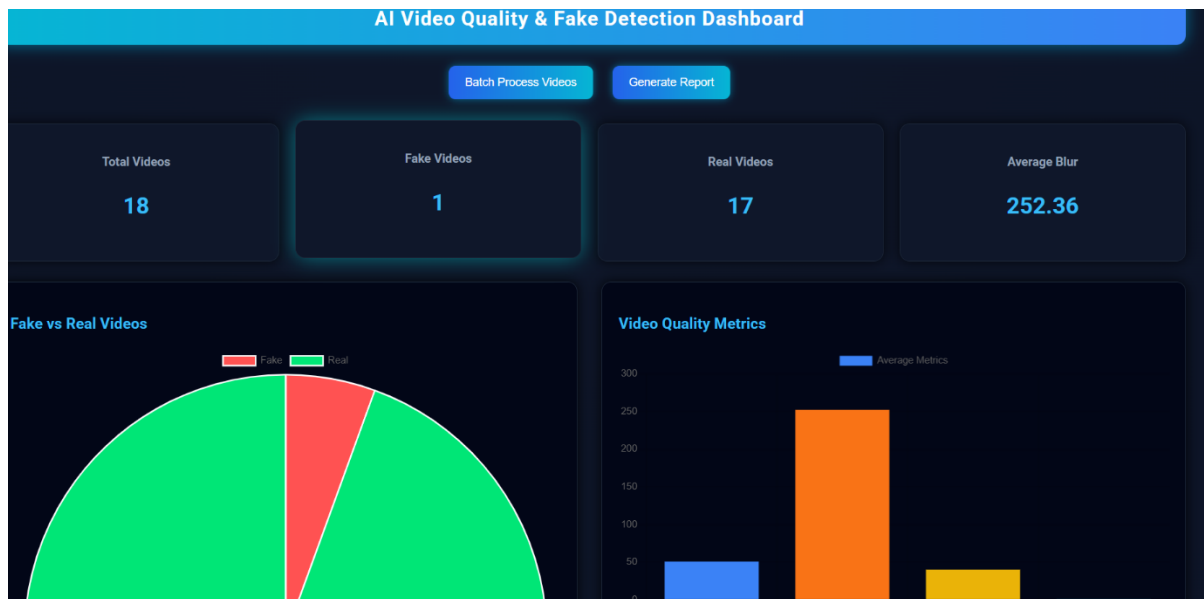
A green button labeled "REAL" is visible next to the text box.

Below the video player is another table titled "WIND Window Analysis (1 windows)".

Window	Blur	Brightness / Energy	Noise	Motion	Quality
1	153.31	66.26	12.37	3.92	Good

A3: VIDEO DASHBOARD (Continued)

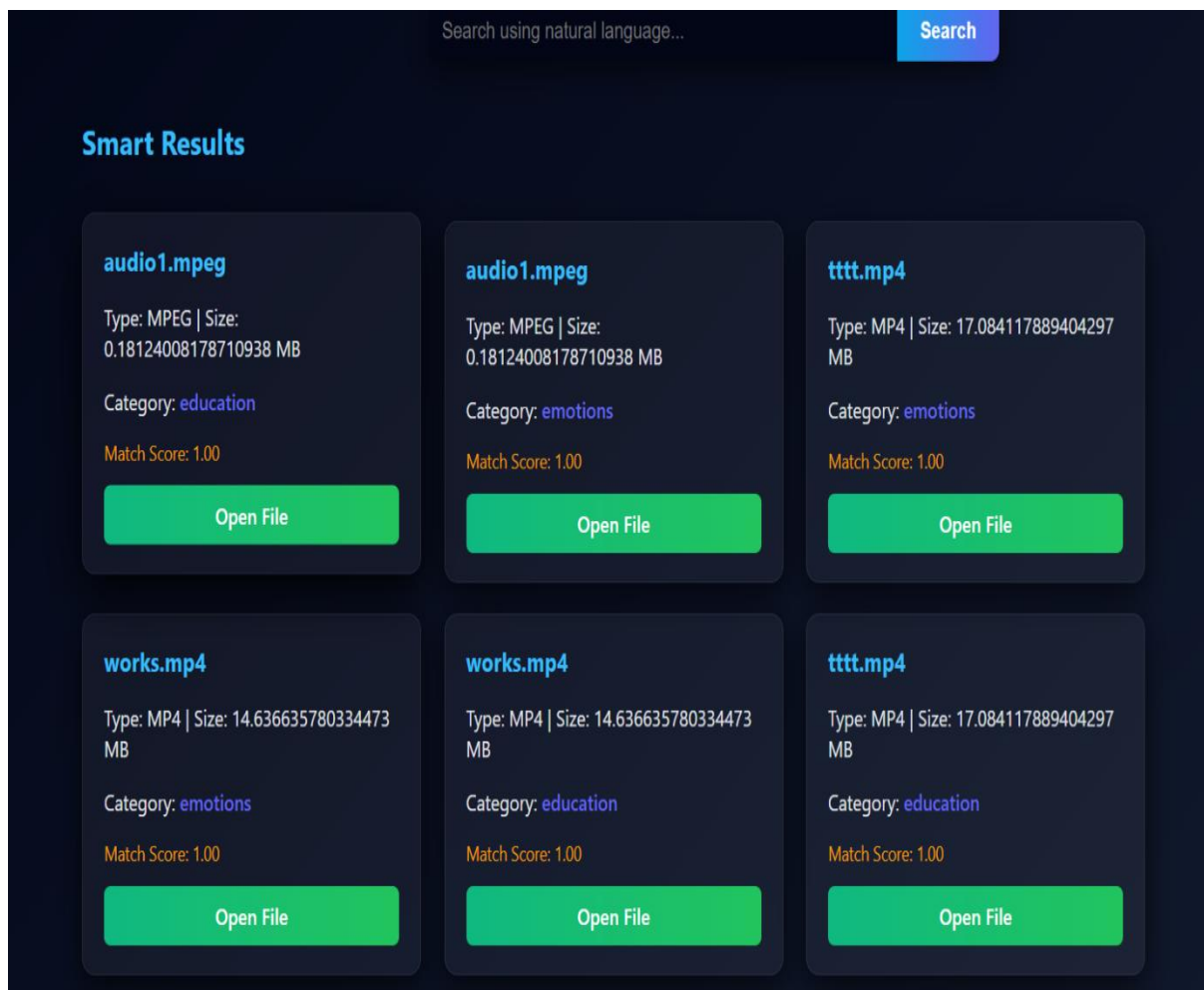
This page displays detailed analysis results of a selected video. It includes resolution, FPS, duration, faces detected, and fake/real prediction. AI metrics like accuracy, WER score, and inference time are also shown. It provides complete insights into video processing.



Duration	Resolution	FPS	Brightness	Blur	Noise	Motion	Audio	Transcript	Prediction	Accuracy	Suggestions
8.28 s	1280x720	90000.0	45.66	276.86	54.74	1.0	1.0	The greatest pleasure I've had is trying to do something that everybody says could not be done.	REAL	-1600.0%	Video is blurry → Use a better camera or adjust focus Low brightness → Increase lighting or adjust exposure High noise → Reduce background noise or improve mic quality
37.67 s	928x922	30.0	68.95	880.76	66.68	1.0	1.0	But the idea was the philosophy behind it that it didn't define me. And even my father, like my grandmother used to always say, oh my god, she's an actress, oh my god, she can't even cook. She'll never get married. And my dad was like, yeah, it's okay, I'll send a cook with her wherever she goes. I was like, yeah, daddy. And he was true till the day he died. I see, I had a cook with me that was preselected by my father. Both my parents were very progressive in their mindsets, especially coming from a country at that time which India was, which was a little bit like this, you know, a woman's life sort of ends when she gets married. My mother still...	REAL	-12600.0%	Video is blurry → Use a better camera or adjust focus High noise → Reduce background noise or improve mic quality

A4: SEARCH PAGE

This page allows users to search for uploaded videos using keywords or filters. It helps quickly locate specific files based on name or analysis results. The search improves usability and saves time. Results are displayed instantly.



A5: UPLOAD PAGE:

This page enables users to upload video or document files for analysis. It validates file type, size, and quality before processing. Users receive feedback messages for successful or failed uploads. This is the entry point for data input.

File Upload System

Search files...

Maximum File Size (MB)

5

Select File Format

PDF

Choose File

Choose file No file chosen

Select Decoding Method

WIND

Reference Text (Ground Truth)

Enter expected speech text for WER comparison

Upload

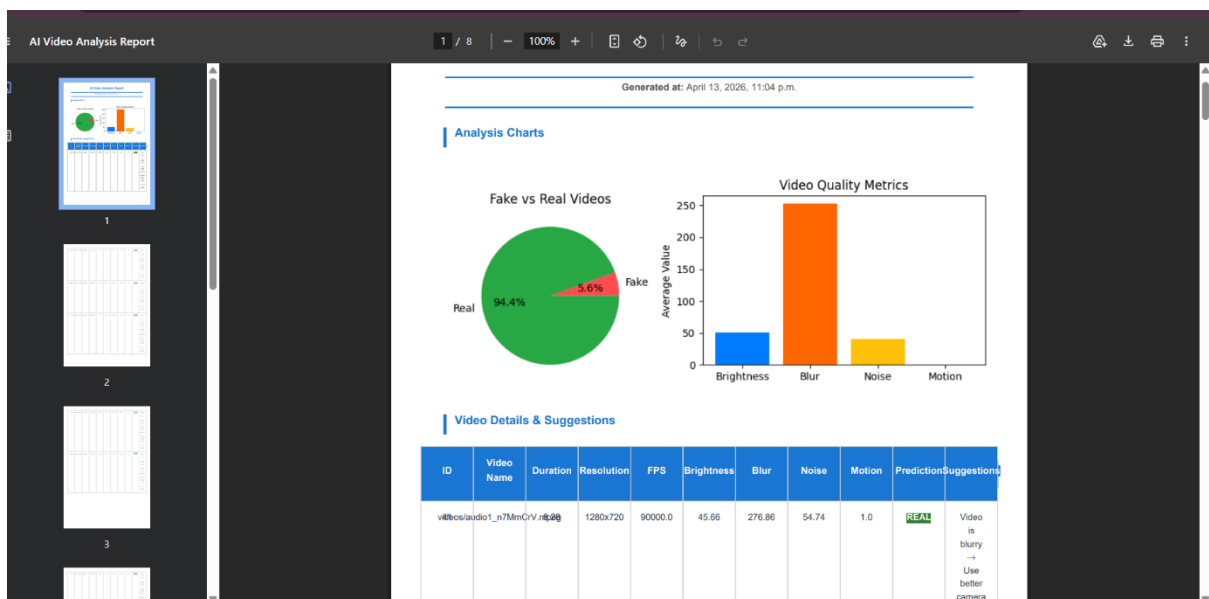
☒ Local
 ☐ Cloud

Do you want to continue? ☒ Yes ☐ No

Uploaded Files

A6: REPORT PAGE:

This page generates a detailed report of the analyzed video. It includes transcript, fake detection result, quality metrics, and performance scores.



REFERENCE

REFERENCE:

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4. "Ian Goodfellow, Yoshua Bengio, & Aaron Courville (2016).
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